**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Algorithms: Examination Techniques

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How do IDE error diagnostics help?

2. Which type suits a count of pupils?

A) Integer
B) Real
C) Boolean
D) Character

3. "A character data type can hold an entire arbitrary-length sentence as one character." Correct this. Explain why.

4. A bill is quantity x unit price + fee. "Calculations always run left to right." Correct this. How does operator precedence affect the result?

## Today's objectives

Explain the core ideas of algorithms: examination techniques using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Read algorithm details**

Distinguish a stated loop boundary, index and test from what the algorithm is intended to do.

**Trace faithfully**

Record each update and branch, including false conditions and early stops.

**Identify prerequisites**

Explain when searching and sorting methods are valid.

**Link method to marks**

An exam explanation should specify precise comparisons, movement or changes to stored data.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Read algorithm details** | Distinguish a stated loop boundary, index and test from what the algorithm is intended to do. |
| **Trace faithfully** | Record each update and branch, including false conditions and early stops. |
| **Identify prerequisites** | Explain when searching and sorting methods are valid. |
| **Link method to marks** | An exam explanation should specify precise comparisons, movement or changes to stored data. |

## I Do - One worked example

**Trace linear search for 9 in [4,6,9,2] with zero-based indexes.**

1. Compare index 0: 4 · 9.
2. Compare index 1: 6 · 9.
3. Compare index 2: 9 = 9; stop.
4. Found at index 2 after three comparisons.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace binary search for 14 in sorted [2,6,10,14,18,22,26].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Complete the index to print the Program Counter abbreviation.**

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[...])
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. A linear search seeks 7 in [3,5,7,9]. Give number of checks. [2 marks]

Q3. State a requirement of binary search. [1 mark]

Q4. Explain what one pass of bubble sort guarantees. [3 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Linear search can work on an unsorted list. ☐ **T** ☐ **F**
2. Distinguish a stated loop boundary, index and test from what the algorithm is intended to do. ☐ **T** ☐ **F**
3. Record each update and branch, including false conditions and early stops. ☐ **T** ☐ **F**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** They show errors and often their location, helping the programmer find the cause. ☐

**Q2** A - Integer ☐

**Q3** A character represents one character; a string represents a sequence of characters. ☐

**Q4** Multiplication comes before addition, giving 10 in the original example. Operator precedence affects the result. Use brackets to make the intended order clear. ☐

## Code First - Model

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[2])
```

**Output: PC**

## We Do - Check the model

1. Index 3 found with one comparison. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** 3. ☐

**Q3** Sorted input. ☐

**Q4** After a full ascending pass, the largest unsorted value has reached the right end of that section. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**